

Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

[Author names removed for double-blind review]

[Affiliations and e-mail addresses removed]

Abstract. Surveying a non-convex site with a quadrotor swarm must remain safe and complete when the wind is strong, the obstacles are unknown in advance, and agents fail in flight. We present a three-layer architecture for this task. The high level partitions the region by polygon-clipped centroidal Voronoi tessellation, plans a boundary-following boustrophedon sweep in every cell, and reallocates the unswept lanes of failed agents to their neighbours. The mid level builds an obstacle map from on-board LiDAR and filters every velocity command through a control barrier function quadratic program whose class- $\mathcal{K}$ function follows from the tilt limit and dead time of the vehicle. The low level is a cascaded PID whose gains are synthesised from either the LQR or the H_∞ Riccati equation. Seven quadrotors were flown in ROS 2/Gazebo over one convex and three non-convex regions under combined hazards: Gauss–Markov turbulence ($\sigma = 2.5$ m/s) over a 3.0 m/s mean wind, nine unmapped static obstacles, two moving obstacles, and the loss of two agents during the mission. Seven of the eight missions finished, with 96.9–99.0% final coverage after the loss of 28.6% of the fleet. PID- H_∞ completed all four regions, lowered the RMS lateral sweep error by 16–39% and finished 11–15% sooner than PID-LQR, at 3.9–5.6 times its torque effort. One PID-LQR mission was lost when an agent touched an obstacle, and both controllers grazed the same obstacle in one region; the largest part of the tracking difference is the rejection of the mean wind. Each configuration was flown once, so these are single-run observations.

INTRODUCTION

Multi-rotor swarms shorten aerial surveys by flying several sensors in parallel, but real survey sites are rarely rectangles. L-, U- and cross-shaped campuses, plants and quarries contain re-entrant corners and narrow arms, the wind near the ground is gusty, obstacles are seldom known before take-off, and vehicles fail. Each of these hazards has been addressed in isolation. Centroidal Voronoi tessellation (CVT) with Lloyd’s iteration distributes agents over a region [1–3] and has been extended to non-convex domains [4, 5], including agent dropout and reinsertion [6]. Boustrophedon sweeps are the standard way to cover a cell [7–10], also for aerial visual coverage [11]. Control barrier functions (CBF) enforce safety as a quadratic program (QP) on the control input [12, 13], with reciprocal multi-robot forms [14, 15]. Linear-quadratic and H_∞ synthesis are classical routes to quadrotor attitude and position control [16, 17], and fault tolerance in swarms is an active topic [18].

What is less reported is how these pieces behave *together* on a physically simulated vehicle, where a velocity command cannot be realised instantly, a barrier that assumes it can is violated, and a planner that assumes perfect tracking loses coverage to wind drift. This paper contributes (i) an integrated architecture that runs polygon-aware CVT partitioning, boundary-following boustrophedon sweeps and lane-level fault reallocation above a LiDAR-mapped CBF-QP safety filter whose class- $\mathcal{K}$ function is derived from the identified closed-loop dynamics of the vehicle; (ii) a side-by-side comparison of PID gains synthesised from the LQR and the H_∞ Riccati equations, both on the linear model and inside the full swarm; and (iii) a data-traceable evaluation of eight full missions under combined turbulence, unmapped obstacles, moving obstacles and agent loss, including an analysis of the failures that occurred. Every number reported is computed by a published script from the recorded telemetry.

METHODOLOGY

System Architecture

Figure 1 shows the three layers. They are defined per agent: each agent owns a cell, a lane queue, a safety filter and a flight controller, and the filter of agent i only uses neighbours within 6 m and obstacles within 10 m. In the simulation reported here the high and mid levels of all agents are *hosted* in one ROS 2 node [19] running at 20 Hz, which shares the coverage grid, the obstacle map and failure notices without communication delay; the low level runs as one node per vehicle at 250 Hz on the odometry stream. A distributed deployment of the upper layers is future work.

High Level: Partition, Sweep and Fault-Tolerant Cooperative Coverage

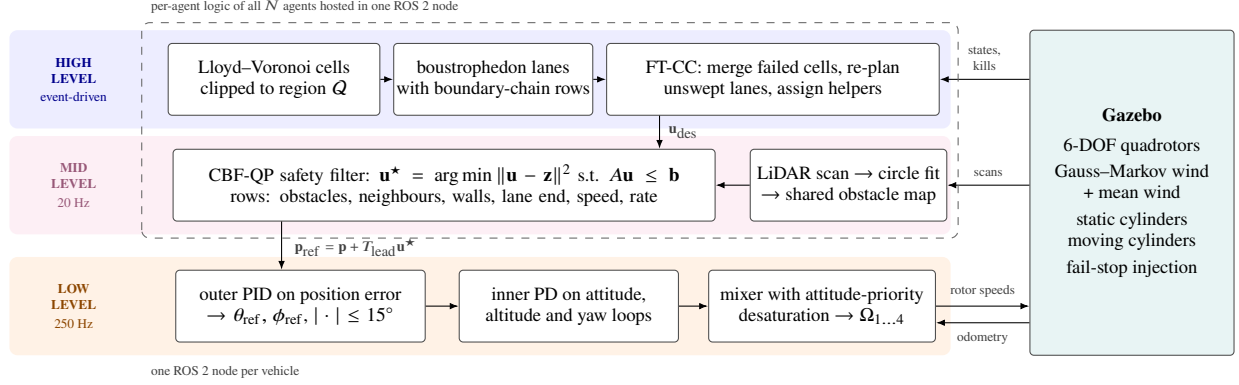


FIGURE 1. Three-layer architecture. Layers are defined per agent; in this study the high and mid levels of all agents are executed in a single ROS 2 node, while each vehicle runs its own low-level controller. The plant, wind, obstacles, sensors and fault injection are simulated in Gazebo.

Given the region polygon Q and generators $\mathbf{g}_i$, the cell of agent i is

$$\mathcal{P}_i = \{\mathbf{q} \in Q : \|\mathbf{q} - \mathbf{g}_i\| \leq \|\mathbf{q} - \mathbf{g}_j\|, \forall j \neq i\}, \quad (1)$$

computed by clipping the bisector cells of the bounding box to Q . Lloyd's iteration $\mathbf{g}_i \leftarrow \text{centroid}(\mathcal{P}_i)$ (25 iterations from deterministic k-means++ seeds) minimises the coverage cost $\sum_i \int_{\mathcal{P}_i} \|\mathbf{q} - \mathbf{g}_i\|^2 d\mathbf{q}$ [3], and agents are assigned to cells by the Hungarian algorithm on their post-take-off positions. Each cell is shrunk by 0.35 m and swept by east–west lanes at most 1.45 m apart. Because the first and last lanes follow the cell's own bottom and top boundary chains rather than a horizontal line, the triangular wedges that a fixed-pitch sweep leaves at slanted Voronoi edges are also flown. With a sensing radius of $r_s = 0.95$ m the swath is 1.90 m, giving at least 24% lateral overlap. Coverage is scored on a 0.30 m grid: a cell inside Q counts once its centre has been within r_s of a live, sweeping agent flying above 0.8 m; cells under static obstacles are excluded from the denominator.

When agent k fails, Fault-Tolerant Cooperative Coverage (FT-CC) (1) keeps the unfinished recovery lanes that k was carrying, (2) erases the coverage recorded inside $\mathcal{P}_k$, treating the data of the lost vehicle as lost, (3) merges all failed cells into $\mathcal{P}_{\text{rec}} = \bigcup_k \mathcal{P}_k$, (4) plans boustrophedon lanes over $\mathcal{P}_{\text{rec}}$ and discards lanes already at least 80% covered, and (5) selects helpers among the neighbours of the failed cells, ranked by distance to the recovery centroid: one helper for a single loss with at most five lanes, otherwise two, and three after a second loss. The lanes are split into contiguous serpentine blocks, one per helper; a helper that has finished its own cell flies to its block directly, otherwise the block is appended to its queue.

Mid Level: Obstacle Mapping and CBF-QP Safety Filter

Obstacle Map

Each 360-ray scan is cleaned of returns within 0.55 m of other agents and 0.80 m of the arena boundary, split into clusters, and each cluster is fitted with a circle. Detections are associated into one swarm-wide track list; a track is confirmed after 12 hits and rejected if it moves. The static obstacle rows of the QP use only confirmed tracks, so the swarm starts with an empty map. The planner deliberately ignores obstacles: lanes are laid over the whole cell and avoidance is left entirely to the safety filter. The two moving obstacles are supplied to the QP from simulator odometry, i.e. they are assumed to be tracked perfectly.

Actuator-Consistent Barrier

With the reference rule $\mathbf{p}_{\text{ref}} = \mathbf{p} + T_{\text{lead}}\mathbf{u}$, the outer PID makes the commanded-velocity response first order with an acceleration limit, $\dot{v} = \text{sat}_{\pm a_{\text{max}}}(k_v(u - v))$, where

$$k_v = gK_d, \quad a_{\text{max}} = g \tan \theta_{\text{max}}, \quad T_{\text{lead}} = \frac{K_d - k_{\text{ff}}}{K_p}, \quad v_c = \frac{a_{\text{max}}}{k_v}, \quad (2)$$

with $\theta_{\text{max}} = 15^\circ$ and feed-forward gain $k_{\text{ff}} = 0.15$ rad/s/m. Evaluated with the LQR outer-loop gains (for both controllers) this gives $k_v = 3.867 \text{ s}^{-1}$, $a_{\text{max}} = 2.631 \text{ m/s}^2$, $T_{\text{lead}} = 0.323 \text{ s}$ and $v_c = 0.680 \text{ m/s}$. The stopping distance

from speed $s > v_c$ is $D(s) = (s^2 + v_c^2)/2a$. Requiring that the distance covered during the dead time T_d plus $D(s)$ fit inside the clearance h yields the admissible approach speed

$$\varphi(h) = \begin{cases} \max\left(0, -aT_d + \sqrt{\max(0, a^2T_d^2 + 2ah - v_c^2)}\right), & h \geq 0, \\ \gamma_r h, & h < 0, \end{cases} \quad (3)$$

with a lumped dead time $T_d = 0.40$ s for sampling, filtering and the attitude loop. It is zero below a finite clearance, a property no linear $\alpha(h) = \gamma h$ has; the branch $h < 0$ ($\gamma_r = 3.5 \text{ s}^{-1}$) demands an outward velocity if the set has been left. The budget a is class dependent, $a = \max(0.15a_{\max}, \mu a_{\max} - |a_{\text{obs}}| - |a_{\text{wind}}|)$ with $\mu = 0.75$ (0.85 for walls) and a wind allowance of 0.5 m/s^2 .

Quadratic Program

For an obstacle or neighbour at $\mathbf{o}$ with unit normal $\hat{\mathbf{n}} = (\mathbf{p} - \mathbf{o})/\|\mathbf{p} - \mathbf{o}\|$, the condition $\dot{h} \geq -\varphi(h)$ is the linear row

$$-\hat{\mathbf{n}}^\top \mathbf{u} \leq \lambda \varphi(h) - \hat{\mathbf{n}}^\top \mathbf{v}_o, \quad h = \|\mathbf{p} - \mathbf{o}\| - (\rho + r_d + \delta), \quad (4)$$

with drone radius $r_d = 0.22$ m, obstacle radius ρ (fitted radius $+0.05$ m for mapped cylinders, 0.50 m for the moving ones), margins $\delta = 0.35$ m (static) and 0.75 m (moving), and $\lambda = 1$ for obstacles. Neighbour rows are reciprocal ($\lambda_{ij} + \lambda_{ji} = 1$, weighted by mission state) and are issued at a hard radius of 0.70 m and a soft radius of 1.20 m. Arena walls and the end of the current lane give half-plane rows $\hat{\mathbf{a}}^\top \mathbf{u} \leq \varphi(h)$, and inscribed polygons bound speed (3 m/s) and the per-tick change of $\mathbf{u}$. Each agent then solves

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \mathbf{z}\|^2 \text{ s.t. } \mathbf{A}\mathbf{u} \leq \mathbf{b}, \quad \mathbf{z} = \frac{\mathbf{u}_{\text{des}} + \mathbf{t} + w \mathbf{u}_{\text{prev}}}{1 + w}, \quad (5)$$

exactly, by enumerating the unconstrained point, the projections onto every row and the pairwise vertices. The tangential bias $\mathbf{t}$ (applied within a 60° cone of a head-on approach) breaks symmetric deadlocks and only enters the objective, never $\mathbf{A}$ or $\mathbf{b}$; $w = 0.25$ smooths successive commands. If the QP is infeasible, soft rows are released (tier 1); if still infeasible, the violation of the hard rows is minimised (tier 2), and a scaled previous command is the last resort (tier 3).

Low Level: LQR and H_∞ Gain Synthesis

Each axis is modelled about hover as a double integrator $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ (e.g. $\mathbf{B} = [0, g]^\top$ from pitch to x , $\mathbf{B} = [0, 1/I_y]^\top$ from torque to pitch) and augmented with an integrator on the input, $\mathbf{x}_a = [\mathbf{x}^\top, u]^\top$. The LQR design solves the algebraic Riccati equation (ARE) for $\mathbf{K}_a = \mathbf{R}^{-1}\mathbf{B}_a^\top \mathbf{P}$. The H_∞ design treats a disturbance w entering through the input channel, $\mathbf{B}_w = [\mathbf{B}^\top, 0]^\top$, and solves

$$\mathbf{A}_a^\top \mathbf{P} + \mathbf{P}\mathbf{A}_a + \mathbf{Q}_a - \mathbf{P}(\mathbf{B}_a \mathbf{R}^{-1} \mathbf{B}_a^\top - \gamma^{-2} \mathbf{B}_w \mathbf{B}_w^\top) \mathbf{P} = 0, \quad (6)$$

rejecting solutions for which $\mathbf{A}_a - \mathbf{B}_a \mathbf{K}_a$ is not Hurwitz [17]. The attenuation level is set to $\gamma = 1.3 \gamma_{\min}$, where $\gamma_{\min}$ is found by bisection. The state-feedback gain is mapped to PID form through the pseudo-inverse of the output-derivative matrix $\Gamma = [C^\top, (CA)^\top, (CA^2)^\top; 0, (CB)^\top, (CAB)^\top]$. The outer loop maps position error to a tilt reference with derivative action on measured velocity, clipped to 15° ; the inner loop is PD on attitude (the synthesised inner integral gain is not used); altitude and yaw loops are designed the same way. A mixer that preserves roll/pitch authority before collective thrust and yaw converts the wrench into rotor speeds. Weights, attenuation levels and the resulting gains are listed in Table 1.

Simulation Setup

Seven 3DR Iris quadrotors ($m = 1.50 \text{ kg}$, $I_{xx} = I_{yy} = 0.0291$, $I_{zz} = 0.0552 \text{ kg m}^2$, arm 0.22 m , $k_f = 8.549 \times 10^{-6} \text{ N s}^2$, rotor speed $\leq 1100 \text{ rad/s}$) are simulated in Gazebo [20] with ROS 2. Controllers read ground-truth odometry; no state estimator is in the loop. Missions start from a launch pad south of the region, cruise at 2.0 m and sweep at 1.6 m/s . Four regions inside a $28 \times 28 \text{ m}$ box are used (Fig. 2): *rect* (784 m^2), *L-shape* (588 m^2), *U-shape* (624 m^2 , 10 m wide notch) and *plus* (460 m^2 , 10 m wide arms), each with nine vertical cylinders ($r = 0.40 \text{ m}$, 4 m tall) placed inside the region. All missions use the same combined-hazard scheme:

TABLE 1. Gain synthesis (outer x/y and inner roll/pitch loops) and linear analysis of the outer loop with the inner loop closed. $\|G\|_\infty$: peak gain from a disturbance acceleration to position; PM/GM: phase and gain margins of the outer loop.

Design	Loop	Q, R	$\gamma_{\min} \rightarrow \gamma$	K_p	K_i	K_d	$\ G\ _\infty$ (s ²)	PM	GM
PID-LQR	outer	diag(0.08, 0.35), 1	–	0.757	0.283	0.394	0.158	37.4°	11.5 dB
	inner	diag(6, 0.6), 0.05	–	4.868	–	0.534			
PID- H_∞	outer	diag(0.10, 0.40), 1	2.54→3.30	2.705	1.092	1.107	0.040	25.2°	9.0 dB
	inner	diag(4, 0.4), 0.06	2.54→3.31	11.604	–	1.189			

TABLE 2. Mission results. T : mission time; lat.: RMS lateral sweep error; $\bar{v}$: mean sweep speed; alt.: RMS altitude error; tilt: 95th percentile while sweeping and maximum; effort: mean $\tau_x^2 + \tau_y^2$ (N²m²); d_{obs} : minimum centre-to-surface distance to a static cylinder; d_{v2v} : minimum inter-agent distance; tier: share of QP solves that needed tier 1 or 2. Killed agents: rect 4, 7 (H_∞) and 7 (LQR); L-shape 4, 7; U-shape 4, 5; plus 1, 7. The PID-LQR rect statistics cover only the 78.8 s before the abort.

Region	Controller	Status	Cov. (%)	T (s)	lat. (cm)	$\bar{v}$ (m/s)	alt. (cm)	tilt ₉₅ (°)	tilt _{max} (°)	effort	d_{obs} (m)	d_{v2v} (m)	tier (%)
rect	PID- H_∞	done	98.2	260	7.2	1.20	2.5	14.7	21.1	0.054	0.49	1.71	2.9
	PID-LQR	abort@78.8	n/a	n/a	11.7	1.12	7.3	14.7	155.9	0.024	0.06	1.52	n/a
L-shape	PID- H_∞	done	98.4	204	8.4	1.19	2.5	14.5	20.8	0.058	0.57	1.32	3.8
	PID-LQR	done	98.5	236	12.2	1.11	2.6	14.5	26.3	0.010	0.29	0.79	4.2
U-shape	PID- H_∞	done	98.5	205	8.3	1.21	2.6	14.8	20.7	0.044	0.63	1.77	4.1
	PID-LQR	done	96.9	240	11.4	1.07	2.5	14.2	16.6	0.011	0.42	1.73	4.1
plus	PID- H_∞	done	98.4	198	9.7	1.15	5.4	14.7	82.5	0.053	0.24	1.24	4.1
	PID-LQR	done	99.0	222	11.5	1.02	2.6	14.4	40.1	0.014	0.24	1.12	4.9

- *Wind.* Each horizontal axis follows a first-order Gauss–Markov process ($\sigma = 2.5$ m/s, correlation time 0.5 s, seed 42), a Dryden-like surrogate [21, 22], plus a constant -2 m/s mean on both axes from $t = 5$ s; the vertical component is scaled by 0.4. The wind is spatially uniform and applied through the Gazebo wind-effects plugin. The recorded wind had mean $(-2.34, -1.95)$ m/s and standard deviation $(2.45, 2.51)$ m/s, with horizontal peaks of 12.0 m/s.
- *Moving obstacles.* Two kinematic cylinders ($r = 0.45$ m) oscillate along the region diagonals with amplitude 10 m (12 m, offset by 1 m, in *plus*) at 0.15 and 0.11 rad/s, i.e. peak speeds of 1.6–2.6 m/s.
- *Agent loss.* When coverage first reaches 18% and 32%, one pre-selected agent is killed (rotor speeds set to zero, fail-stop) and the coordinator is notified at once. Victims were chosen to own comparable cells next to the launch pad (Table 2).

Both controllers use the same planner, filter and plant model; Eq. (2) is evaluated with the LQR outer-loop gains in both cases. Each region–controller pair was flown once ($N = 1$) and recorded on video. The two *plus* missions also include three transit guards added before they were flown (a stall detector, a speed reduction above 16° tilt and a 20 s start-wait timeout). Metrics are computed from 20 Hz controller logs, discarding the rows after an agent is killed and the first 20 s. Lateral error is the deviation from the lane reference during steady east–west sweeps (heading stable for 1.5 s, speed above 0.6 m/s); torque effort is the time average of $\tau_x^2 + \tau_y^2$; obstacle distance is measured from the drone centre to the cylinder surface. The collision geometry of the airframe reaches 0.235 m (body) to 0.348 m (rotor tips) from its centre, so a surface distance below these values means certain or possible contact.

RESULTS AND DISCUSSION

Mission Completion and Fault Recovery

Table 2 and Fig. 2 summarise the eight missions. Seven finished with every surviving agent back at its cell centroid and final coverage between 96.9% and 99.0%, although two of seven agents (28.6% of the fleet) were lost at 18% and 32% coverage. In every mission the obstacle map contained all nine cylinders, with centre and radius errors within the 0.1 m resolution of the log, plus at most two confirmed tracks that matched no cylinder; no completed mission needed a tier-3 QP fallback. Figure 3 shows the recovery mechanism: each kill causes a small drop, because the coverage inside the failed cell is erased, after which the helpers resume the original slope; Fig. 4 shows the resulting coverage maps. The PID-LQR U-shape mission ended just below the 97% target (96.9%) when all agents had exhausted their lanes; its final coverage map shows thin uncovered strips along the boundaries between neighbouring cells in the left arm, which are nearly absent in the PID- H_∞ map of the same region. We did not isolate their cause. The PID-LQR rect mission was aborted at 78.8 s and is analysed below.

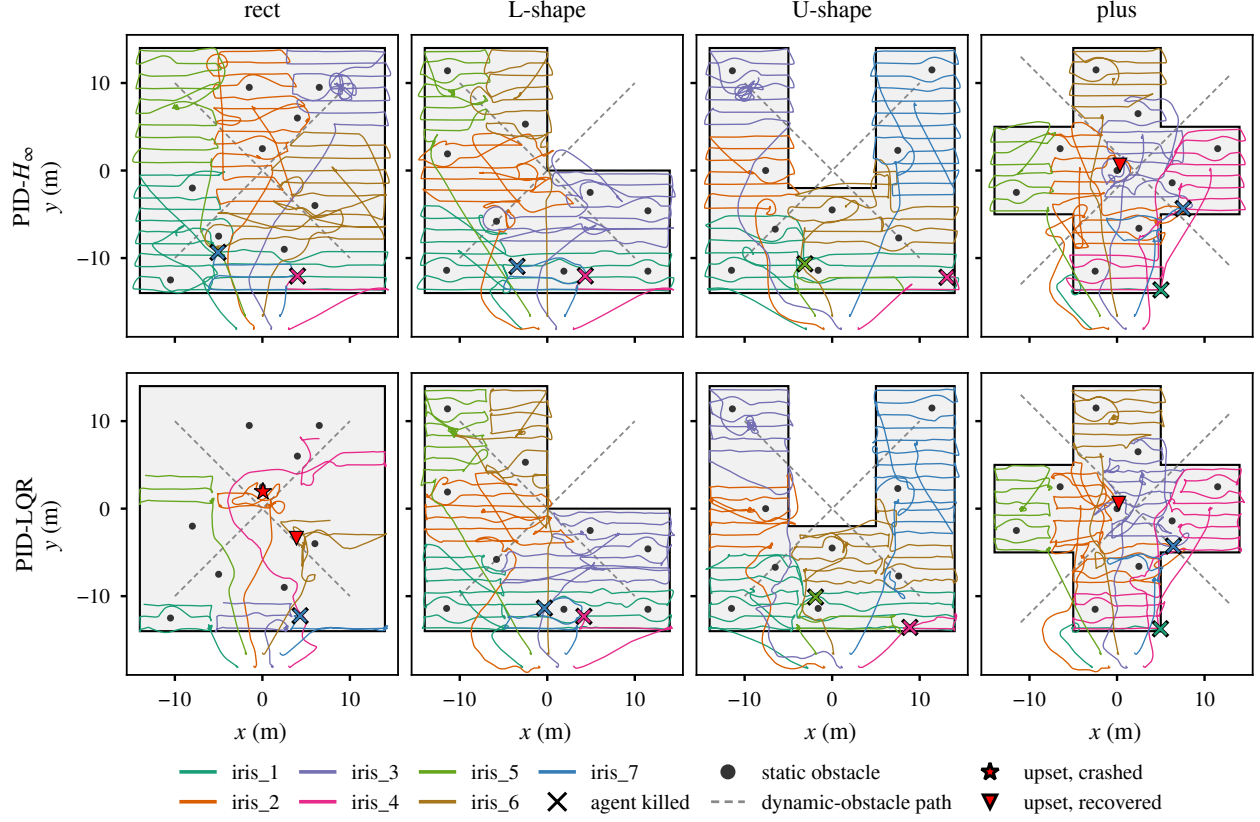


FIGURE 2. Flown trajectories (live agents above 0.5 m) for all eight missions, including transit from the launch pad at $y \approx -18$ m. Dark discs: static cylinders; dashed: nominal paths of the moving cylinders; crosses: position where an agent was killed; red markers: tilt upsets above 30° .

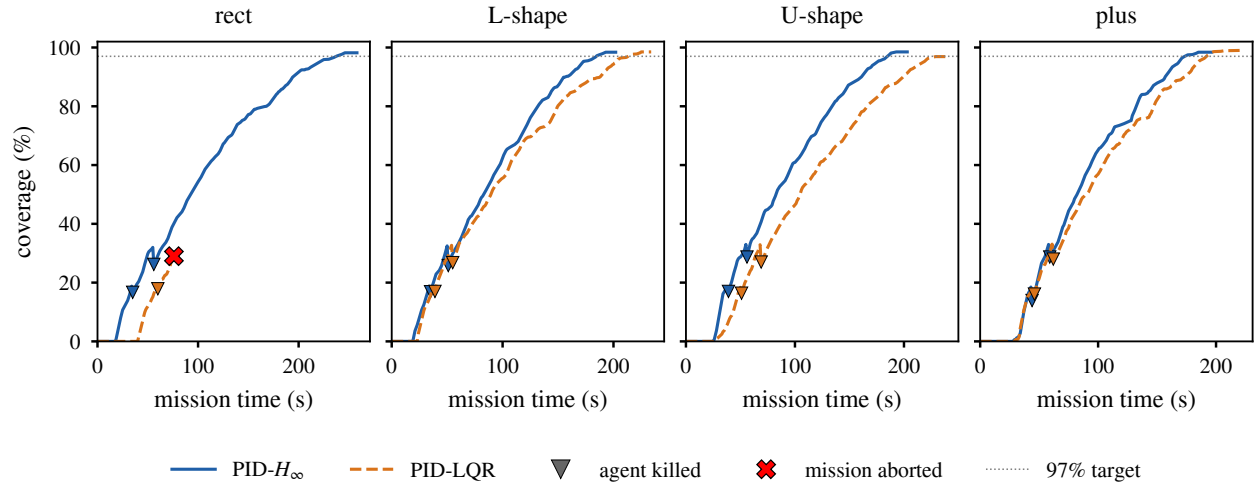


FIGURE 3. Coverage against mission time (coordinator clock, 1 Hz). Triangles mark agent kills; the step after each kill is the erased coverage of the failed cell being re-swept.

Tracking under Turbulence: LQR versus H_∞ Gains

The two designs differ clearly in how tightly they hold a lane (Fig. 5a). The median lateral error was 2.7–3.8 cm with PID- H_∞ and 8.5–8.8 cm with PID-LQR, and the RMS error fell from 11.4–12.2 cm to 7.2–9.7 cm, a 16–39% reduction that holds in all four regions. Splitting the error into its mean and its fluctuation shows where the difference lies. PID-LQR flew with a downwind offset of -5.7 to -7.0 cm in every region (the mean wind blows towards $-y$), whereas

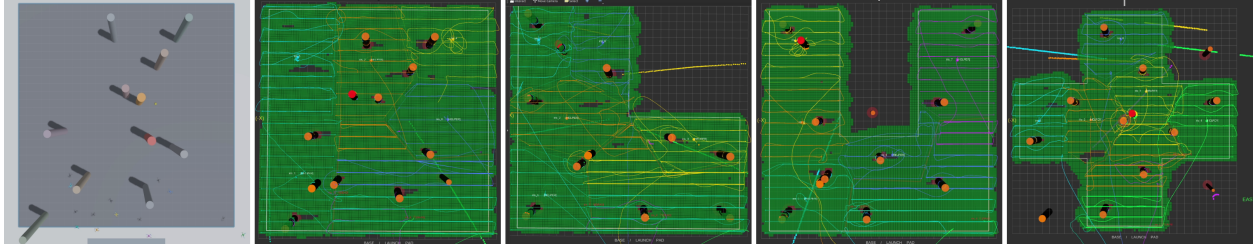


FIGURE 4. (a) Gazebo view of the rect mission; (b)–(e) RViz coverage maps (green) at the end of the PID- H_∞ missions in rect, L-shape, U-shape and plus, with lanes, flown paths and mapped obstacles. Frames are extracted from the mission recordings.

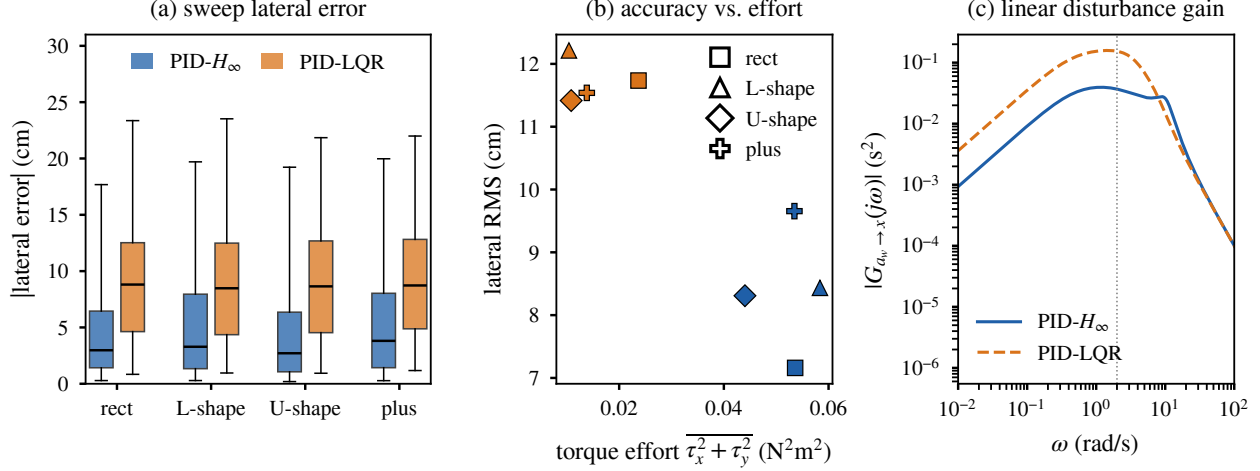


FIGURE 5. (a) Absolute lateral error during steady east–west sweeps (box: quartiles, whiskers: 5th–95th percentiles). (b) RMS lateral error against torque effort per mission. (c) Gain from a disturbance acceleration to position of the linearised outer loop; dotted: corner frequency of the wind model.

PID- H_∞ stayed within 0.7–1.6 cm of the lane; the standard deviation differed less (7.0–9.6 vs 9.4–10.8 cm). The offset is a low-frequency effect: the H_∞ outer integral gain is 3.9 times larger, and the PID-LQR node additionally freezes its position integrator whenever the error exceeds 0.25 m, which was the case in 36–45% of its sweep samples. The mean sweep speed was 1.15–1.21 m/s against 1.02–1.12 m/s for a 1.6 m/s command, which accounts for the 11–15% shorter missions where both finished. Altitude error and the 95th-percentile tilt were essentially equal, since both are bounded by the same tilt limit. The linear analysis in Table 1 explains the direction of the result: the H_∞ gains lower the worst-case gain from disturbance acceleration to position by a factor of 4.0 and the steady-state position deviation under the Gauss–Markov input by 3.7 (Fig. 5c).

The price is visible in the same table. PID- H_∞ used 3.9–5.6 times the torque effort of PID-LQR in the completed pairs (Fig. 5b), although motors were saturated in less than 0.2% of samples in every mission, and its outer loop has a higher crossover (9.08 vs 4.09 rad/s) with smaller phase and gain margins. The measured error reduction is also far smaller than the linear factor, because the tilt limit, the reference pre-filter, the rate limits of the QP and the wind force model are not in the linear model. Two further caveats limit the attribution: the synthesis weights Q and R are not identical in the two designs, and the integrator gating described above belongs to the PID-LQR node, not to the synthesis. The comparison is therefore between two complete gain designs; this study does not isolate the contribution of the attenuation level γ from that of higher loop gain.

Safety: Clearance, Contact and Upset Events

No two agents came closer than 0.79 m (PID-LQR, L-shape) and all minima stayed above the 0.70 m hard radius, so there was no inter-agent contact. Against static cylinders the minimum surface distance under PID- H_∞ was 0.49, 0.57 and 0.63 m in rect, L-shape and U-shape, clear of the airframe, while under PID-LQR it was 0.06, 0.29 and 0.42 m. In *plus* both controllers passed within 0.24 m of the central cylinder, inside the rotor envelope. Figure 6 shows the upsets that followed, and the coordinator log shows a common pattern: each contact happened while the agent was stepping from one lane to the next beside a cylinder, and in each case the QP had run in tier 2 for 1–4 s with that

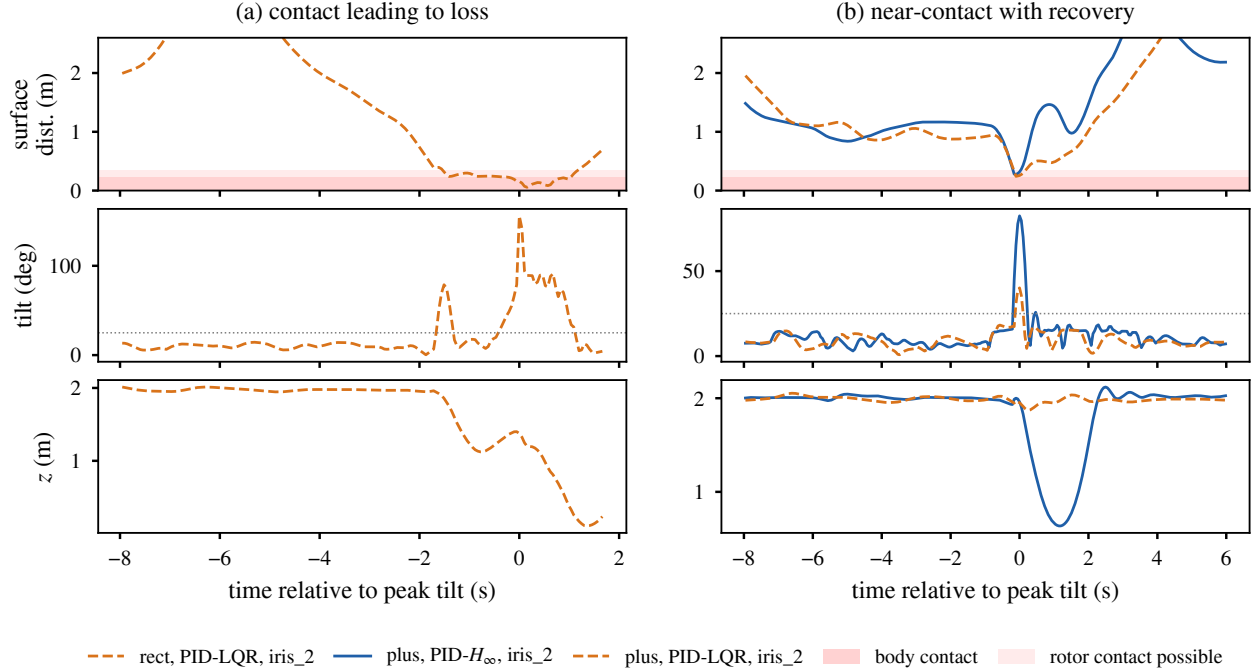


FIGURE 6. Upset events, aligned at peak tilt. Top: centre-to-surface distance to the nearest static cylinder (dark band: body contact, light band: rotor contact possible); middle: tilt (dotted: 25° level-out threshold); bottom: altitude. (a) PID-LQR, rect, agent 2: contact and loss. (b) *plus*, agent 2 under both controllers: grazing contact and recovery.

cylinder as the binding row. In the PID-LQR rect mission, agent 2 turned into its third lane next to the cylinder at (0, 2.5), reached a surface distance of 0.06 m, i.e. body contact, tilted to 155.9° and fell; the crash detector aborted the mission. In *plus*, agent 2 of both missions stepped from its fourth to its fifth lane next to the central cylinder: under PID-LQR the tilt reached 40.1° with little height loss, under PID- H_∞ it reached 82.5°, the vehicle dropped to 0.64 m and regained cruise altitude within about 3 s. With one run per case, none of these events can be attributed to the controller. They are better explained by the design choice that lanes are planned without obstacle knowledge, which lets lane ends and steps fall inside the safety margin of a cylinder, combined with a barrier built on a 0.22 m disc and a fixed 0.5 m/s² wind allowance that gusts of up to 12.0 m/s exceed. A further 50.8° upset (PID-LQR, rect, agent 6 at 24.4 s) occurred in transit away from any cylinder and was recovered. Moving obstacles also entered their 0.75 m margin: the QP logged negative barrier values against them in 8 of the eight missions, the lowest (−0.68 m) corresponding to a centre-to-surface distance of 0.34 m, at the rotor envelope. Because these values are only logged while the QP runs in tier 2, they bound the true minimum from above. A collision envelope matching the rotor tips, obstacle-aware lane ends and a gust-dependent acceleration budget are the obvious remedies.

Limitations

The results rest on one run per configuration with a fixed wind seed, perfect state feedback, perfectly tracked moving obstacles, instantaneous failure detection, a geometric coverage model without camera or photogrammetric error, and centralised execution of the upper layers. The planner and barrier components were not ablated, and no coverage method from the literature was re-implemented for comparison. Statistical claims, estimator and communication effects, and a distributed implementation remain to be addressed.

CONCLUSION

A three-layer architecture combining polygon-clipped Voronoi partitioning, boundary-following sweeps, lane-level fault reallocation, a LiDAR-mapped actuator-consistent CBF-QP and Riccati-synthesised PID control was evaluated in eight simulated missions with seven quadrotors under strong turbulence, unmapped and moving obstacles, and the loss of two agents. Seven missions finished with 96.9–99.0% coverage. Gains synthesised from the H_∞ Riccati equation held the lanes 16–39% more tightly and shortened missions by 11–15% compared with LQR gains, at several times the torque effort and with smaller stability margins; both designs suffered obstacle contacts in gusts, one of which ended

an LQR mission. Future work will repeat the campaign over many wind seeds, enlarge the collision envelope and add wind to the barrier model, distribute the upper layers onto the vehicles, and move to flight tests with on-board state estimation.

REFERENCES

1. S. P. Lloyd, IEEE Transactions on Information Theory **28**, 129 (1982).
2. Q. Du, V. Faber, and M. Gunzburger, SIAM Review **41**, 637 (1999).
3. J. Cortés, S. Martínez, T. Karatas, and F. Bullo, IEEE Transactions on Robotics and Automation **20**, 243 (2004).
4. A. Breitenmoser, M. Schwager, J.-C. Metzger, R. Siegwart, and D. Rus, in *2010 IEEE International Conference on Robotics and Automation* (2010) pp. 4982–4989.
5. K. Cao, Y. Chen, S. Gao, H. Dang, and D. An, Applied Sciences **13**, 8530 (2023).
6. K. Lee and K. Lee, IEEE Access **12**, 5503 (2024).
7. H. Choset and P. Pignon, in *Field and Service Robotics*, edited by A. Zelinsky (Springer, London, 1998) pp. 203–209.
8. H. Choset, Annals of Mathematics and Artificial Intelligence **31**, 113 (2001).
9. E. Galceran and M. Carreras, Robotics and Autonomous Systems **61**, 1258 (2013).
10. S. Hu, T. Xu, and B. Wang, Sensors **21**, 2221 (2021).
11. S. Papatheodorou and A. Tzes, in *2017 IEEE 56th Annual Conference on Decision and Control (CDC)* (2017) pp. 4662–4667.
12. A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, IEEE Transactions on Automatic Control **62**, 3861 (2017).
13. A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, in *2019 18th European Control Conference (ECC)* (2019) pp. 3420–3431.
14. L. Wang, A. D. Ames, and M. Egerstedt, IEEE Transactions on Robotics **33**, 661 (2017).
15. D. Zhou, Z. Wang, S. Bandyopadhyay, and M. Schwager, IEEE Robotics and Automation Letters **2**, 1047 (2017).
16. M. Belkheiri, A. Rabhi, A. El Hajjaji, and C. Pégard, in *2012 2nd International Conference on Communications, Computing and Control Applications (CCCA)* (IEEE, Marseilles, France, 2012) pp. 1–6.
17. J. C. Doyle, K. Glover, P. P. Khargonekar, and B. A. Francis, IEEE Transactions on Automatic Control **34**, 831 (1989).
18. J. O’Keeffe, IEEE Robotics and Automation Letters **10**, 8954 (2025).
19. S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, Science Robotics **7**, eabm6074 (2022).
20. N. Koenig and A. Howard, in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vol. 3 (Sendai, Japan, 2004) pp. 2149–2154.
21. U.S. Department of Defense, “Flying qualities of piloted aircraft,” Tech. Rep. MIL-HDBK-1797 (U.S. Department of Defense, 1997).
22. R. W. Beard and T. W. McLain, *Small Unmanned Aircraft: Theory and Practice* (Princeton University Press, Princeton, NJ, 2012).